

Lab #01: Registers¹

Overview

uVision Build Test

Task 0: File Set-up → Partners 1&2

Assembly Coding

Task 1: Group Averages (25 pts.) → Partner 1

Task 2: Sum of Squared Errors (25 pts.) → Partner 2

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Task 4: Total Sum of Squares (25 pts.) → Partner 2

Submission (through Canvas Assignments)

Overview:

During producing electronic components, we often want to compare different groups of resistors to see if there is a statistical difference between them on some measure. For more than 2 groups, we often use an analysis of variance. For this lab, you will calculate some of the statistical values that are used in an ANOVA test of statistical significance.

You will be comparing three groups of resistors: 1) 'A' group, 2) 'B' group, and 3) 'C' group. In each group, there are 5 different resistor samples. We will see if there are statistical differences in the resistor values of the three groups of resistors.

The value of each resistor is given to you in the starter project as EQU statements. You MUST use those EQU labels (e.g., ARes1, CRes5) in your code. If the values for those labels change your program must still work correctly.

Use the Excel spreadsheet given to you as a guide. NOTE that some of your answers will differ from the spreadsheet. Why do you think that is?

For more information on the ANOVA statistical test see: [How to Perform a One-Way ANOVA by Hand - Statology](#), or [Analysis of variance - Wikipedia](#).

Task 0: File setup → Partner 1&2

- ❖ Edit the comments at the top of the file to show the correct information. Be sure that BOTH partners' names are listed. Provide a brief description of your program in the comments
- ❖ Assemble (build) the starter file given for this lab. Make sure that you are not receiving errors before you start adding code to the main.s file.

¹ This lab design is adapted from the work by **Dr. Diana De La Rosa-Pohl**.

Task 1: Group Averages (25 pts) → Partner 1

- ❖ Calculate the mean for each resistor group. Place the mean of the A group in **R12**. Place the mean of the B group in **R11**. Place the mean of the C group in **R10**.
- ❖ Calculate the “Grand Mean” which is the average for all resistors and store in **R9**.
- ❖ Here i is the group (A-C), j is the j th resistor in each group, and N is the total number of the resistor in each group, we have

$$\text{Group mean: } \bar{x}_i = \frac{\sum_{j=1}^N x_{ij}}{N}$$

$$\text{Grand mean: } \bar{X} = (\bar{x}_A + \bar{x}_B + \bar{x}_C)/3$$

Task 2: Group Averages (25 pts) → Partner 2

- ❖ Calculate the sum of squared errors (SSE) for each group. Place the result from the A resistor group in **R8**. Place the result from the B resistor group in **R7**. Place the results from the C resistor group in **R6**.
- ❖ Here i is the group (A-C), j is the j th circuit in each group, and N is the total number of the circuits in each group, we have

$$\text{Sum of squared error: } (SSE)_i = \sum_{j=1}^N (x_{ij} - \bar{x}_i)^2$$

Task 3: Regression sum of squares (25 pts.) → Partner 1

- ❖ Calculate the regression sum of squares. Place the result in **R5**.

$$\text{Regression sum of squares: } (SSR) = \sum_{i=A}^C (\bar{x}_i - \bar{X})^2$$

Task 4: Total sum of squares (25 pts.) → Partner 2

- ❖ Calculate the total sum of squared errors $(SSE)_{Total}$ and place the result in **R4**.
- ❖ Calculate the total sum of squares (SST) and place the result in **R3**.

$$\text{Total sum of squared errors: } (SSE)_{Total} = (SSE)_A + (SSE)_B + (SSE)_C$$

$$\text{Total sum of squares } (SST) = (SSR) + (SSE)_{Total}$$

At the end of your program, you should have the following information in your registers:

Register	Value
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R0	Don't care
R1	Don't care
R2	Don't care
R3	Total sum of squares
R4	Total sum of squared errors
R5	Regression sum of squares
R6	Sum of squared errors for C group
R7	Sum of squared errors for B group
R8	Sum of squared errors for A group
R9	Grand mean
R10	Mean of C group
R11	Mean of B group
R12	Mean of A group
R13	RESERVED
R14	RESERVED
R15	RESERVED

Submission (Through Canvas)

- ❖ Submit the complete and working project to Canvas as **ONE ZIP FILE**.
- ❖ The project file will be name Lab1.uvprojx.
- ❖ The project folder will be named ECE_3346_Lab1_Team#
- ❖ The zip file should be named ECE_3346_Lab1_Team#.zip
- ❖ **Both** partners need to submit the complete project in a zip file.